

# Collapse-Causal Agent Tracker (C-CAT) — Seed Note

*Formerly: Actor Tracker — Seed Note*

*Status: queued, not started. Pick up after a few more general (effect-layer) trackers are built.*

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## A Note on the Rename

This project was first sketched as the “Actor Tracker.” It is renamed here to the Collapse-Causal Agent Tracker (C-CAT), for two reasons, not one cosmetic one.

First, the rename makes explicit a relationship that the old name left implicit: this tracker is the layer directly beneath the collapse-causal tracker (C-CT), described in its own companion document, RICD Collapse-Causal Tracker: What It Is. C-CT measures the aggregate, second-order effects a cause leaves behind in a system’s own trajectory, without needing to know what that cause was. C-CAT is the tracker built to answer the question C-CT leaves open: what, specifically, produced this. Matching the names lets the two documents read as one stack instead of two unrelated projects.

Second, “actor” undersold half of what this project turned out to need. It implies a person, a decision-maker who can be interviewed, deposed, or held accountable. But the underlying question, what specifically drove a given collapse, applies just as much when the answer is a mechanism rather than a decision: a tumor’s mutation cascade, a soil system’s feedback loop, a reef’s thermal-stress response. “Agent” covers both without strain — a state actor and a chemical agent are both agents, and nobody confuses the two — in a way “actor” never could.

## The Core Distinction This Project Is Built On

Three layers, not two:

1. Mechanism — the actual causal actions of an identifiable agent, human or non-human (a specific administrator diverting funds; a specific office deciding which grants to cancel and when; a specific mutation cascade; a specific land-use pattern driving topsoil loss). This layer is never observed by any tracker built so far.
2. Second order effect, functioning as general cause — the aggregate, measurable consequence those actions leave behind (a resource trajectory that stays flat while adaptive capacity erodes; a debt-spike hitting several institutions simultaneously; a resource inflow smaller than expected by a specific, unexplained dollar amount). This is what every tracker built so far actually measures, and it is the real, correct causal level for predicting a flip — the flip responds to this layer, not to the underlying actions directly.
3. Flip — the classified outcome itself.

Existing terminology in the framework (extraction flux, push-vectors, the propagation machinery) already lives at layer 2, the effect layer. Naming the earlier project “actor tracker” rather than “vector tracker” was a deliberate choice to avoid blurring that distinction — “vector” is already doing real work one layer up from where this project lives. Naming this version “collapse-causal agent tracker” rather than a shorter “agent tracker” is the same kind of choice:

“collapse-causal” is already claimed, precisely, by the tracker one layer up, and reusing it here rather than coining an unrelated name keeps the two documents legible as one stack.

## What This Project Actually Is

Not a data-science extension of the existing trackers. A genuinely different methodology from C-CT — but one that itself splits into two methodologies running in opposite directions, depending on whether the agent in question is a person or a mechanism.

### Human Agents

Closer in practice to forensic accounting or investigative journalism than to the IPEDS-style structured-data work C-CT is built on. The existing effect-layer trackers work because their data (IPEDS, Form 990s, bond filings) is effect-layer by construction: institutions are required to report outcomes on a regular schedule, regardless of whether anything went wrong.

A human-agent tracker needs mechanism-layer evidence instead: board minutes, deposition transcripts, whistleblower testimony, subpoenaed correspondence, OIG and state AG investigative findings. This kind of evidence is:

- Adversarial to obtain — no actor voluntarily files a report describing their own predatory action the way an institution files an annual finance survey.
- Fragmentary by nature — it typically exists at all only because something already went wrong and someone investigated after the fact, not on a predictable reporting cycle.
- Scale-appropriate to a research program, not a solo project; my own estimate, stated directly: this would require a whole graduate program’s worth of researchers and investigators to assemble properly, not an individual effort.

### Non-Human Agents

Where the agent is a mechanism rather than a person, the same three-layer structure applies, but the evidentiary problem runs in the opposite direction from the human-agent branch. A mechanism doesn’t conceal its own action from an observer, and it isn’t the subject of an investigation that exists only because something already went wrong — entire scientific fields (oncology, soil science, marine biology, epidemiology) exist specifically to measure and quantify these mechanisms on an ongoing basis, independent of whether collapse is underway. Some working examples: cancer progression (tumor registries, histopathology, the broader oncology literature), desertification (satellite and soil-moisture monitoring, soil-science literature), and coral reef collapse (reef-temperature and bleaching-survey programs, marine biology literature).

Where the human-agent branch needs a whole research program’s worth of forensic-accounting-style investigators, the non-human branch is closer to a literature-review and data-sourcing problem: identifying which already-running instrumentation program covers a given mechanism and drawing from it, rather than building an investigation from nothing.

Two of the working examples above — desertification and coral reef collapse — complicate this cleanly, though, and deliberately: both have real human components (land-use decisions, emissions policy) sitting alongside the non-human mechanism. Those cases won’t sort neatly

into one branch or the other; they are the cases where a C-CAT adapter has to hand off between the human-agent and non-human-agent methodology, or run both and reconcile them. That handoff is an open design question, not yet worked out (see Open Questions, below), not a flaw in the two-branch framing ; rather it is exactly where the two branches need to talk to each other.

## What the Tracker Would Actually Be Looking For

Not generic evidence that “a decision was made somewhere” or “something happened somewhere.” Specifically: evidence of the named mechanisms RICD already formalizes mathematically but that the effect-layer trackers built so far can only ever infer indirectly, through their downstream consequences. Confirmed examples from the manuscript itself:

- Frantic Striking Phase (§8.8b) — the search-and-triage cost pattern of repeated, failing survival campaigns.
- Lateral knocks / vented heat (§8.6a, H<sub>i,j</sub>) — an unmetabolized failure discharged onto a neighboring container.
- Voluntary Mutual Sacrifice (§10.5b.11b.iii, “sacrificial lifeboat” in working shorthand) — two comparably-struggling containers where the discharge asymmetry resolves in favor of one’s survival through the other’s own trajectory redirection, distinct from predatory targeting (§10.5b.11b.i) or accidental capture via declining connectivity (§10.5b.11b.ii).

These are agent-neutral: RICD’s container dynamics don’t specify whether the agent instantiating a given mechanism is a person or a process, so the same named mechanisms apply on either branch of C-CAT, e.g. a lateral knock can be a hospital system offloading its own distress onto a neighboring provider, or a stressed reef offloading thermal load onto an adjacent one through changed water chemistry.

C-CT and its kind — collapse-causal trackers, or equivalently downstream consequence trackers — measure the aggregate effect these mechanisms leave behind without ever needing to identify which named mechanism, or which kind of agent, produced it. C-CAT’s whole purpose is the piece that process leaves out: given a real, observed collapse, was it actually Frantic Striking, a lateral knock, a sacrificial-lifeboat trade, predatory targeting, accidental capture, or a non-human mechanism running the equivalent pattern; a real, falsifiable attribution question the effect-layer data was never built to answer.

## What “Done” Would Look Like, Eventually

Two deliverables, matching the pattern already used for C-CT, but the shape of each deliverable differs sharply between the two branches.

### Human-Agent Branch

1. A prospective adapter — real fields, real sourcing requirements, scoped to what forensic-layer evidence can actually supply (case files, not CSVs).
2. A how-to manual — but a genuinely different kind of manual than the one written for C-CT. Not “how to extract field F2A18 from a CSV.” Closer to: how to build a case file; what

counts as evidence of a specific decision versus merely being suggestive of one; how to establish provenance and reliability for fragmentary, adversarially-obtained sources; and how to avoid quietly sliding a layer-2 (effect) finding into a layer-1 (mechanism) claim it doesn't actually support.

## **Non-Human-Agent Branch**

1. A prospective adapter scoped to whatever instrumentation program already covers the mechanism in question, pulling from an existing reef-monitoring or soil-moisture dataset, for instance, rather than building new evidentiary infrastructure from scratch.
2. A how-to manual of a different kind again, not how to establish provenance for adversarially-obtained evidence, but how to avoid the opposite failure: treating an abundant, well-instrumented mechanism-level dataset as if it already confirms a collapse-level effect, when the two need to be checked against each other independently, the same way C-CT and C-CAT need to be checked against each other on the human-agent branch. Abundance of data is not the same thing as confirmation of the specific causal claim.

## **A Real, Concrete Case Worth Using as a Working Example**

### **Human Agent: Abundant Blessings**

A Hyde Park charity whose executive director was charged in a real, documented \$23 million federal fraud case (HUD-OIG and FBI investigation), built on fabricated vendor invoices using stolen real company names and logos, and self-dealing rent payments disguised as market-rate third-party leases. The detail that matters most for this project specifically: when a LAHSA investigator asked whether the charity's board was aware of how money was being spent, it turned out the board itself was fabricated: some listed members did not exist, others had never heard of the organization or its director.

This surfaces a real complication C-CT never had to face. External Governance Attestation (RICD Definition 6.1.1) worked there because the attesting bodies — accreditors, AAUP — are genuinely independent of the institution being assessed, and their findings are a matter of public record regardless of what the institution itself says. The human-agent branch of C-CAT cannot assume this. In a case like Abundant Blessings, the nominal oversight structure is itself part of the fabrication, not a real, independent check on it. Any human-agent adapter would need its own way of verifying that a claimed governance or oversight structure actually exists and is actually independent, before treating its attestations as trustworthy, a genuinely new kind of validation problem, not a variant of anything C-CT needed.

A sharper way to state this (RICD 10.2, flagged there as conceptual, not yet tested against data): a container's real boundary is constituted by wherever agent energy is actually interfacing with a recognizable pattern — its energetic signature — not by any official claim about that boundary. Ordinary institutional decline is the signature drifting from real activity slowly enough that the gap eventually leaks into a container's own downstream numbers, which is what let C-CT catch real decline years early without ever observing an agent directly. Fraud is different in kind, not degree: it is the manufacture of a signature specifically engineered to have no real container behind it, built so an outside observer mistakes the claim for the fact; Abundant Blessings'

fabricated board is a clean instance. The working hypothesis for why the human-agent branch specifically needs mechanism-layer evidence, stated here as something to test rather than something already confirmed: ordinary drift leaks into effect-layer data on its own, but a signature engineered to fake its own container is built to suppress exactly that leak, which is plausibly why effect-layer data alone will not be enough once fraud, rather than ordinary decline, is the actual phenomenon.

This concealment dynamic is specifically a human-agent problem, worth stating plainly now that there is a non-human branch to contrast it with. A mechanism cannot engineer a fake signature the way a fraudulent human agent can; a tumor's histology doesn't lie about what it is, a reef's bleaching signature isn't fabricated to fool an inspector. So the mechanism-layer evidence a non-human-agent adapter needs doesn't face the "is this attestation genuinely independent" problem the human-agent branch has to solve for cases like Abundant Blessings. That is a second, independent reason — beyond raw data abundance — that the two branches sit at opposite ends of the difficulty spectrum: the human branch has to solve concealment as its own problem; the non-human branch mostly does not have concealment to solve.

### **Non-Human Agent: No Working Case Yet**

Unlike Abundant Blessings, nothing here has been checked against a specific real, documented instance, and none should be invented for a seed note. When this branch is picked up, the first real task is finding a documented case where a mechanism-level cause is independently confirmed against a real collapse or effect-layer signal, ideally one where the mechanism data predates or independently corroborates the trajectory, the same evidentiary relationship External Governance Attestation gave the higher-ed tracker on the human side.

### **Open Questions for Later**

- Whether either branch's classifier target is even the same kind of thing as a collapse/flip classification, or whether it is structurally a different task entirely (e.g., attributing a known effect to a specific agent, rather than predicting an outcome from features). Worth thinking through once real candidate case material is in hand, not from first principles now.
- How a mixed-causation system (desertification, coral reef collapse) should route between the human-agent and non-human-agent methodology: a single adapter handling both, or an explicit handoff protocol between two separate ones.
- For the non-human branch specifically: what independent evidentiary relationship, parallel to External Governance Attestation, would let a mechanism-level finding bypass or calibrate the effect-layer classifier the way a governance verdict does on the human branch (or perhaps the non-human branch doesn't need an override mechanism at all, precisely because it doesn't face engineered concealment); this question needs answering.